

# Failure-Aware Mode Switching for End-to-End Autonomous Driving

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Detecting when a fast driving policy is about to fail — and when to escalate to a slow VLA

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# The problem: fast policies fail where it matters

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- **Fast E2E policy**
  - real-time, drives by default — but degrades on rare, long-tail situations
- **Slow VLA model**
  - reasons robustly about hard cases — but too slow / heavy to run continuously
- **The dilemma**
  - can't run the heavy model always; can't trust the fast model on the hard 1%
- **Key question: when should we switch from the fast policy to the slow one? (System 1 → System 2)**

# Related work

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## Hybrid E2E + VLA (dual-system)

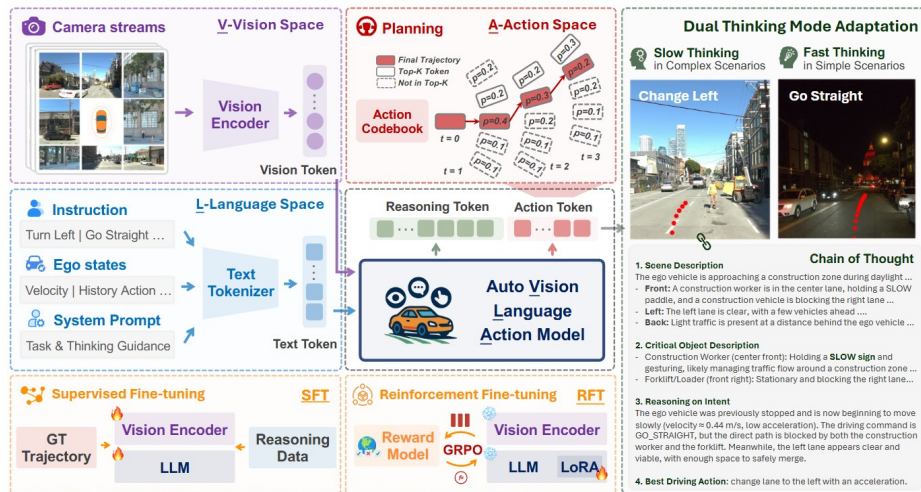
- AutoVLA — fast/slow reasoning via RL
- FASIONAD — switch on an uncertainty threshold
- AdaDrive — learns when to invoke the LM
- Senna / DriveVLM-Dual — a VLM guides an E2E planner

*Most learn an internal switch tied to one architecture.*

## Uncertainty & failure prediction

- Deep ensembles; max-softmax OOD baselines
- **OOD  $\neq$  failure — novelty doesn't imply error**
- **Our angle: model-agnostic — ask which observable signal best predicts a measured failure, so a trigger sits on top of any planner.**

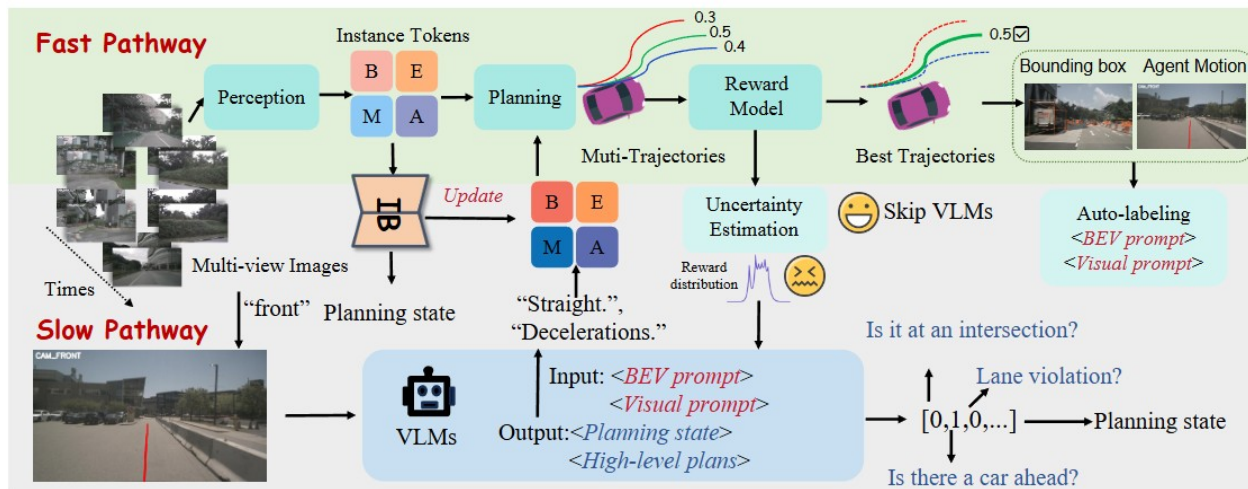
# Related work — AutoVLA: a dual fast/slow VLA model



**Architecture** — one Vision-Language-Action model; emits a chain-of-thought (slow) in complex scenes or a direct action (fast) in simple ones; trained by SFT then RL fine-tuning (GRPO).

**Our switching view** — the fast/slow switch is baked into one model and learned end-to-end; ours is a model-agnostic trigger on top of any planner (no retraining) that detects failure, not 'reason-more'.

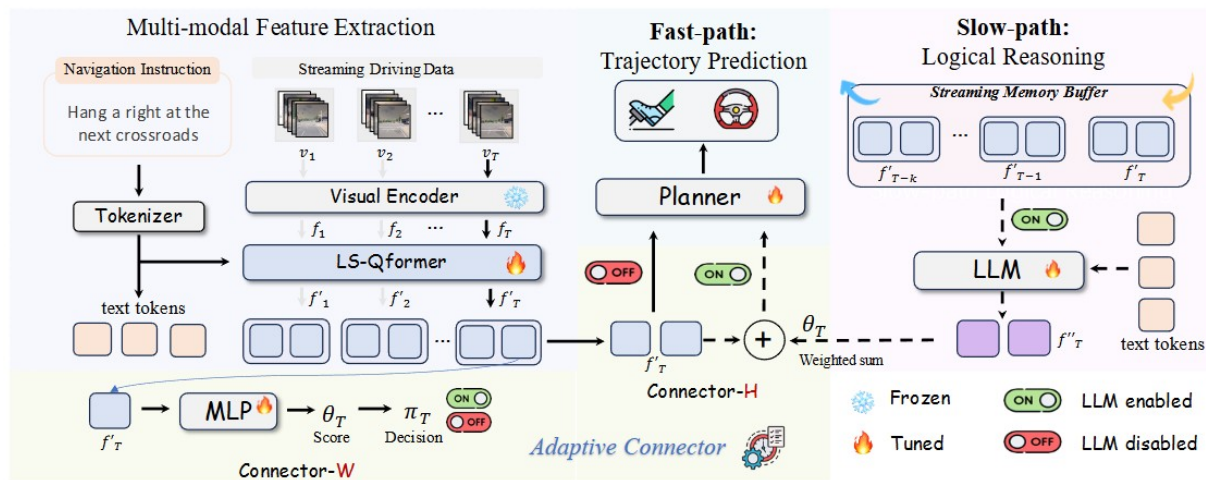
# Related work — FASIONAD: uncertainty-gated fast/slow fusion



**Architecture** — fast pathway (perception → planning → reward model → best trajectory) + a slow VLM pathway answering structured questions; reward-distribution uncertainty decides when to call vs skip the VLM.

**Our switching view** — its trigger is the model's own uncertainty (a single-model signal). We find native uncertainty weak (0.55–0.61) but cross-model disagreement strong (0.76–0.82) — disagreement may be the better trigger.

# Related work — AdaDrive: a learned gate for the slow LLM



**Architecture** — frozen encoder + tuned Q-former; an Adaptive Connector learns WHEN to switch the LLM on/off (Connector-W) and HOW to fuse it (Connector-H); a fast planner plus a slow LLM over a streaming memory.

**Our switching view** — its gate is learned jointly and tied to the architecture; it doesn't say which signal carries failure information or whether it transfers. We identify the signal (cross-model disagreement) and test transfer.

# Heightened attention is triggered by uncertainty



Unpredictable behavior



Unclear intent



Degraded perception



Negotiation needed

A driver's shift shows in gaze & pulse — a model's shows in **cross-model disagreement**, the signal our detector reads.

# Scope: the detector only

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- This work studies the detector, not the full switch.
- A passive monitor — scores each frame for impending failure; does not change the trajectory or invoke the VLA.
- Why — isolates alarm quality from driving performance, and makes signals comparable across very different models.
- Inference-only; closed-loop switching is future work.



# Method: measured failure, four candidate signals

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## Failure label (measured, not opinion)

- **Top-10% trajectory error (ADE) = failure**
- Measured against logged ground truth
- **ADE is never an input feature → no circularity**
- Scene-level cross-validation → no leakage

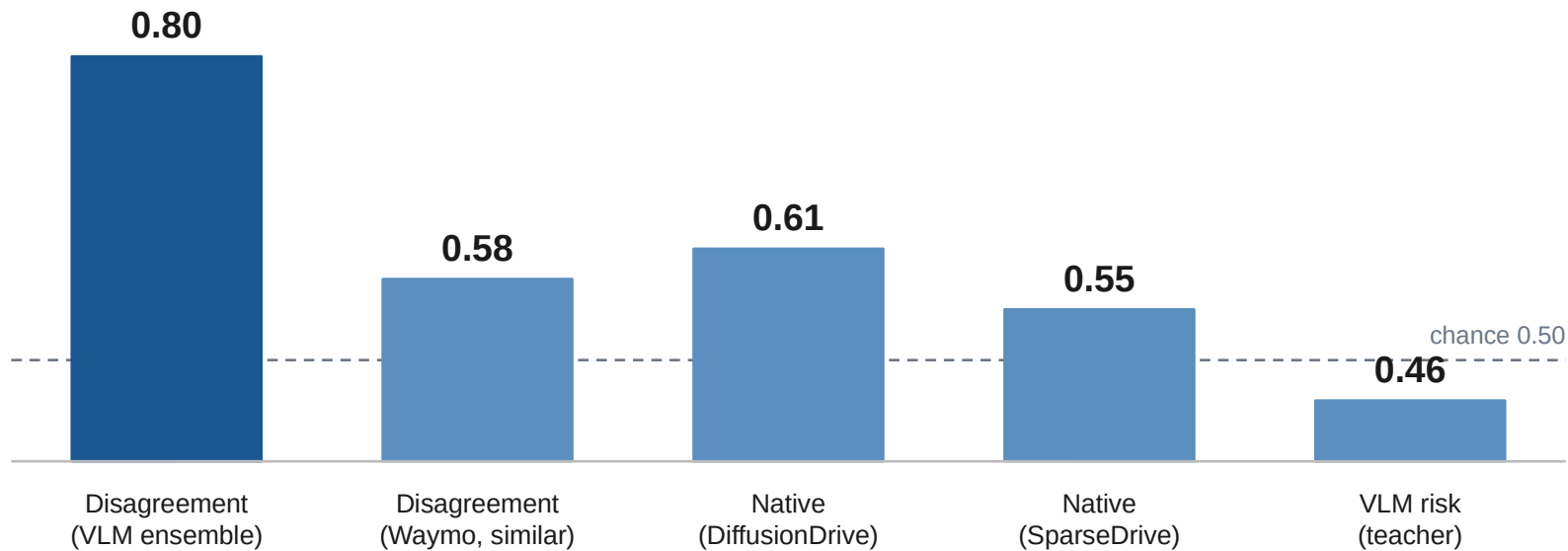
## Four candidate signals

- ① Cross-model disagreement
- ② Single-model native uncertainty
- ③ VLM-perceived risk
- ④ Temporal accumulation (CUSUM / EWMA)

# Which signal predicts failure?

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- Cross-model disagreement is by far the most informative failure signal



AUROC (0.50 = chance, 1.00 = perfect). Failure = top-10% ADE.

# A learned detector reaches AUROC $\approx 0.82$

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- Predictive power lives in cross-model disagreement, not a model's self-assessment

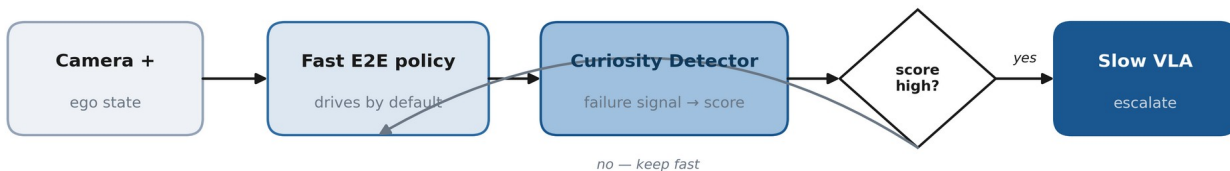
| Detector                   | Input                     | AUROC          |
|----------------------------|---------------------------|----------------|
| Logistic regression        | 8 disagreement statistics | 0.815          |
| MLP (64, 32)               | 8 disagreement statistics | 0.812          |
| DeepSets                   | raw trajectories          | 0.819          |
| Single-model internal only | one model's own signals   | $\approx 0.61$ |

All three learned models are statistically indistinguishable ( $\approx 0.82$ ) on the disagreement signal; **restricting to a single model's internal signals drops to  $\approx 0.61$ .**

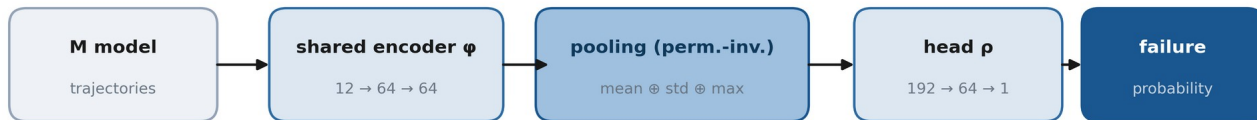
# Architecture: a DeepSets detector over model trajectories

- Permutation-invariant std / max pooling directly encodes cross-model disagreement

(a) Fast-slow driving with a failure trigger



(b) Curiosity Detector — DeepSets over the set of model trajectories



*permutation-invariant std / max pooling directly encodes cross-model disagreement*

# The signal transfers to a real driving planner

- Disagreement predicts SparseDrive's actual failure — even with SparseDrive excluded

| Detector (predicting SparseDrive's failure) | AUROC        | AUPRC        |
|---------------------------------------------|--------------|--------------|
| Native only (SparseDrive's own signals)     | 0.618        | 0.170        |
| Disagreement only (SD, DD, VAD)             | 0.755        | 0.290        |
| Fused (disagreement + native)               | 0.762        | 0.293        |
| <b>Leave-SD-out (DD, VAD → SD failure)</b>  | <b>0.667</b> | <b>0.191</b> |

- Official nuScenes-val, 6019 frames; label = SparseDrive top-10% ADE; 5-fold scene CV.
- Decisive: even excluding SparseDrive, disagreement still predicts its failure (0.67) → genuine transfer, not self-reference.**
- Conservative — only 3 similar driving models; a more diverse ensemble would be higher.

# What drives it: near-term, speed disagreement

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- Failures coincide with disagreement about the immediate move, not distant geometry

| Feature (logistic-regression coefficient) | Weight |
|-------------------------------------------|--------|
| Path-length disagreement                  | +1.37  |
| 1 s waypoint spread                       | +0.98  |
| 3 s waypoint spread                       | −0.79  |

*Coefficients are illustrative — the disagreement statistics are correlated, so individual weights should not be over-interpreted.*

# What doesn't work

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- **VLM-perceived risk  $\neq$  failure**
- AUROC 0.46 ( $\approx$  chance). Scene-danger is agent-agnostic; failure is agent-relative (OOD  $\neq$  failure).
- **Temporal accumulation — no gain**
- single-frame 0.80 > EWMA 0.76 > CUSUM 0.72; open-loop failures are single-shot, nothing to accumulate.
- **A real fast-slow model (AutoVLA)**
- the trigger applies, but that checkpoint's slow $\approx$ fast quality makes it a weak testbed  $\rightarrow$  future work.

# Qualitative: what the hard scenes look like

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- Real nuScenes scenes from the top-10% error band (failures).
- Hard, ambiguous conditions (night, unclear layout) — high mean ADE, a majority of frames fail.
- **Exactly where the models disagree and the detector fires.**



# Discussion

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- **Why disagreement works**
- when plausible interpretations diverge, models err — the computational echo of conflict-monitoring in human attention.
- **Why VLM-risk fails**
- scene-danger is agent-agnostic; failure is agent-relative. Disagreement reflects the models' own confusion.
- **Efficiency → distillation**
- running an ensemble defeats the purpose; distill the signal into one cheap forward pass (the deployable form).

# Limitations & future work

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## Limitations

- 0.82 is measured within the VLM ensemble; transfer (0.76 / 0.67) addresses this, but official val used only 3 driving models.
- Disjoint sets — strong VLM signal and native signals not fully fused.
- Open-loop replay lacks sequential dynamics.
- Ensemble is costly — a research probe, not a deployment design.

## Future work

- **Distill disagreement into a single-pass trigger.**
- Closed-loop evaluation (Bench2Drive / NAVSIM).
- Validate on a real nuScenes fast–slow model (decision-level protocol).
- Add the remaining cognitive signals (surprise, novelty, stakes).

# Conclusion

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- Cross-model disagreement is the strongest failure signal — 0.80 (VLM ensemble), 0.76 for a real planner (0.67 with the target excluded), vs 0.62 native.
- A learned detector (incl. permutation-invariant DeepSets) reaches 0.82; single-model internal only 0.61.
- VLM-perceived risk and temporal accumulation don't help — OOD  $\neq$  failure; open-loop is single-shot.
- **Path forward: distill the signal into a cheap trigger, and evaluate in closed loop.**

Thank you · Questions?